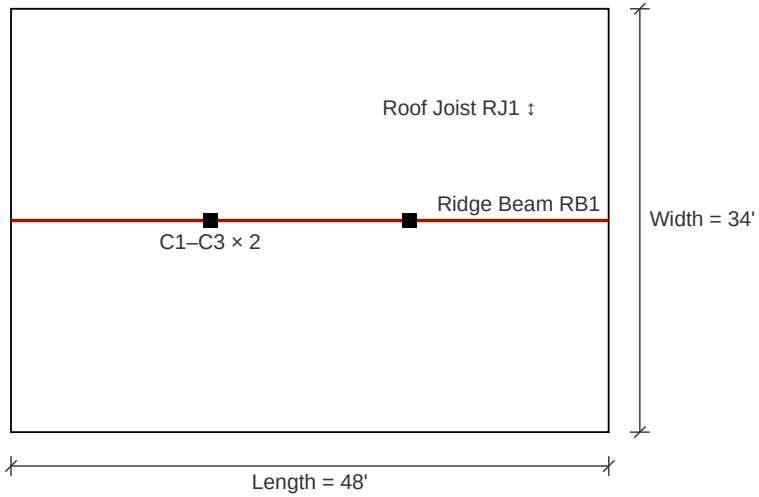
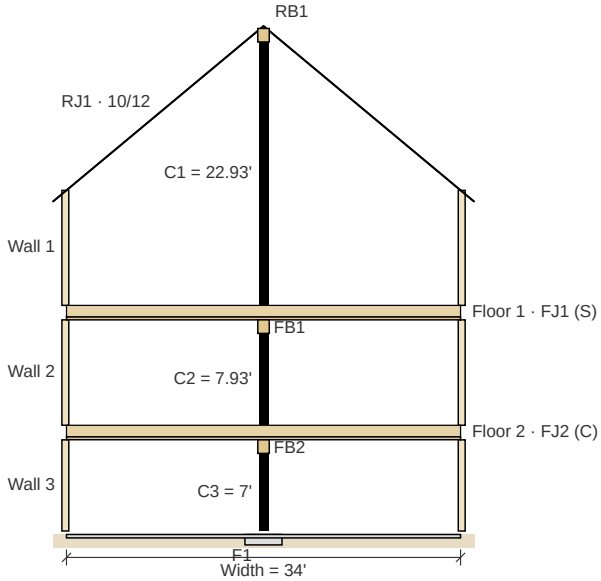
Test Review – F24 (Wawa) — worked key

Plan · beam spans 48' ÷ 3 = 16' between columns



Section through the main house (at the columns)



Location

City	**Wawa**

Roof 1

Pitch	**10**	/12
Overhang	**14**	inches
Dead load	**15**	psf
Number of interior columns	**2**	
RB1: Simple or continuous	**C**	S or C

Floor 1

Live Load	**40**	psf
Dead Load	**15**	psf
Joists: Simple or continuous	**S**	S or C
Directly applied ceiling	**Y**	Y or N
Beam: Simple or continuous	**C**	S or C

Floor 2

Live Load	**40**	psf
Dead Load	**25**	psf
Joists: Simple or continuous	**C**	S or C
Directly applied ceiling	**N**	Y or N
Beam: Simple or continuous	**C**	S or C

General Information

Width	**34**	feet
Length	**48**	feet
Wall 1 height		inches
Wall 2 height		inches
Wall 3 height		inches
Subfloor thickness	**0.750**	inches
Soil capacity	**110**	KPa

Assumptions: 1. Walls have 0" width. 2. All beams and columns are weightless. 3. Footings sized on Total Factored Loads. Column heights are given on the form.

Test Review – F24 (Wawa) — worked key

Snow Loading

Ground snow load S_s	3.4	kPa
Associated rain load S_r	0.4	kPa
Specified snow load	2.27	kPa

0.55 × 3.4 + 0.4 = **2.27 kPa**

Specified snow load	47.44	psf

2.27 × 20.9 = **47.44 psf**. His key keeps **2 decimals** here.

Roof Joists RJ1

Unfactored snow load	47.44	psf
Unfactored dead load	15	psf
Pitch	10	/12
Maximum span	17	feet

Walls are 0" wide, so the span is the length to the ridge — half the width, 34 ÷ 2: **17'**

Overhang	14	inches
Joist depth	16"	inches

S = 47.44 psf → **50LL + 15DL** column. 10/12 is **High** slope (Low ≤ 6/12). Need ≥ 17' = 17'-0". Any spacing/depth/series in that column whose span is long enough is right.

Joist type	s31	
Spacing	24"	inches
Slope length	283.77	inches

(17 × 12 + 14) × √(1 + (10/12)²) = 218 × 1.3017 = **283.77"** (p.21 slope factor 1.302 gives the same within ½")

D factor	13.33	inches

16 × 10 ÷ 12 = **13.33"** (his key). Table p.21 says 13⅜" — both accepted.

Length to order	25	feet

(283.77 + 13.33) ÷ 12 = 24.76' → round up = **25'**

Floor Joist FJ1

Unfactored live load	40	psf
Unfactored dead load	15	psf
Subfloor thickness	0.75	inches
Directly applied ceiling	Y	Y or N
Simple or continuous span	S	S or C
Span	17	feet

Walls 0" wide, beam in the middle: 34 ÷ 2 = **17'**

Joist depth	9½"	inches

p.6: 40 LL + 15 DL, 3/4" OSB, directly applied ceiling, "simple or continuous" columns. Need ≥ 17'-0".

Joist type	s47	
Spacing	16"	inches
Length to order	17	feet

walls 0" wide: the span itself, rounded up to a whole foot = **17'**

Floor Joist FJ2

Unfactored live load	40	psf
Unfactored dead load	25	psf
Subfloor thickness	0.75	inches
Directly applied ceiling	N	Y or N
Simple or continuous span	C	S or C
Span	17	feet

Walls 0" wide, beam in the middle: 34 ÷ 2 = **17'**

Joist depth	9½"	inches

p.7: 40 LL + 25 DL, 3/4" OSB, no directly applied ceiling, "continuous" (either table works — continuous-only spans are longer) columns. Need ≥ 17'-0".

Joist type	s47	
Spacing	16"	inches
Length to order	34	feet

one continuous joist over the beam: 17 + 17 (the full width) = **34'**

Ridge Beam RB1

Span	16	feet

Length ÷ (2 columns + 1) = 48 ÷ 3 = **16'**

Tributary width	17	feet

Width ÷ 2 = 34 ÷ 2 = **17'** — half of each joist span lands on the beam.

Unfactored snow load	807	plf

47.44 × 17 = 806.48 → round up = **807**

Unfactored dead load	255	plf

15 × 17 = 255 → round up = **255**

Unfactored total load	1,062	plf

807 + 255 = **1,062**

Total factored load	1,530	plf

1.5 × 807 + 1.25 × 255 = 1529.25 → round up = **1,530**

Beam choice	LVL 5¼×16	

Use the **16.5'** row (round 16' up). Need LL ≥ 807, TL ≥ 1,062, TFR ≥ 1,530. Any product/size that passes all three is right (LSL p.6–7, LVL p.8–9, PSL p.10–11).

Beam height	16	inches
Bearing requirements	8.2	inches

Bottom row of the table for your beam: **3.3" end / 8.2" intermediate**. Beam is continuous over the columns → intermediate is the one that matters; both are accepted.

Floor Beam FB1

Span	16	feet

Length ÷ (2 columns + 1) = 48 ÷ 3 = **16'**

Tributary width	17	feet

Width ÷ 2 = 34 ÷ 2 = **17'** — half of each joist span lands on the beam.

FJ1: Simple or continuous	S	S or C
Multiplier for S or C span	1	

Simple joists: **1**

Unfactored live load	680	plf

40 × 17 = 680 → round up = **680**

Unfactored dead load	255	plf

15 × 17 = 255 → round up = **255**

Unfactored total load	935	plf

680 + 255 = **935**

Total factored load	1,339	plf

1.5 × 680 + 1.25 × 255 = 1338.75 → round up = **1,339**

Beam choice	LVL 5¼×14	

Use the **16.5'** row (round 16' up). Need LL ≥ 680, TL ≥ 935, TFR ≥ 1,339. Any product/size that passes all three is right (LSL p.6–7, LVL p.8–9, PSL p.10–11).

Beam height	14	inches
Bearing requirements	6.4	inches

Bottom row of the table for your beam: **2.6" end / 6.4" intermediate**. Beam is continuous over the columns → intermediate is the one that matters; both are accepted.

Floor Beam FB2

Span	16	feet

Length ÷ (2 columns + 1) = 48 ÷ 3 = **16'**

Tributary width	17	feet

Width ÷ 2 = 34 ÷ 2 = **17'** — half of each joist span lands on the beam.

FJ2: Simple or continuous	C	S or C
Multiplier for S or C span	1.25	

Continuous joists put 25% more on the beam: **1.25**

Unfactored live load	850	plf

40 × 17 × 1.25 = 850 → round up = **850**

Unfactored dead load	532	plf

25 × 17 × 1.25 = 531.25 → round up = **532**

Unfactored total load	1,382	plf

850 + 532 = **1,382**

Total factored load	1,940	plf

1.5 × 850 + 1.25 × 532 = 1940 → round up = **1,940**

Beam choice	LVL 5¼×16	

Use the **16.5'** row (round 16' up). Need LL ≥ 850, TL ≥ 1,382, TFR ≥ 1,940. Any product/size that passes all three is right (LSL p.6–7, LVL p.8–9, PSL p.10–11).

Beam height	16	inches
Bearing requirements	8.2	inches

Bottom row of the table for your beam: **3.3" end / 8.2" intermediate**. Beam is continuous over the columns → intermediate is the one that matters; both are accepted.

Interior Column C1

RB1: Simple or continuous	C	S or C
Multiplier for S or C span	1.25	

Continuous beam puts 25% more on the column: **1.25**

Total factored load	30,600	pounds

RB1 TFL 1,530 × span 16' × 1.25 = 30600 → round up = **30,600 lb**

Column height	22.93	feet

Given on the sheet: **22.93'**

Column choice	None works	

Round 22.93' up to the next row; need ≥ 30,600 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right. **Nothing in the table works**: the biggest, a 7" × 7" Parallam, carries 23,945 lb at that height. Ask Mark what he wants written (steel column, or more columns).

Interior Column C2

FB1: Simple or continuous **C** S or C

Multiplier for S or C span **1.25**

Continuous beam puts 25% more on the column: **1.25**

Total factored load **57,380** pounds

C1 30,600 + FB1 1,339 × 16 × 1.25 (26,780) = **57,380 lb** — C2 holds up C1 too.

Column height **7.93** feet

Given on the sheet: **7.93'**

Column choice **PSL 5¼×7**

Round 7.93' up to the next row; need ≥ 57,380 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right.

Interior Column C3

FB2: Simple or continuous **C** S or C

Multiplier for S or C span **1.25**

Continuous beam puts 25% more on the column: **1.25**

Total factored load **96,180** pounds

C2 57,380 + FB2 1,940 × 16 × 1.25 (38,800) = **96,180 lb** — everything above lands here.

Column height **7** feet

Given on the sheet: **7'**

Column choice **PSL 7×7**

Round 7' up to the next row; need ≥ 96,180 lb. TimberStrand stops at 14', Parallam goes to 24'. Any that passes is right.

Column Footing F1

Total factored load on C3 **96,180** pounds

Soil capacity **110** KPa

Soil capacity **2299** psf

110 × 20.9 = **2299 psf**

Footing area **41.84** square feet

96,180 ÷ 2299 = **41.84 ft²**

Length of square footing **77.62** inches

√41.84 = 6.468' × 12 = **77.6"**

Smallest dimension of column **7** inches

C3 is 7" × 7" Parallam 1.8E → smallest side **7"**

Thickness of square footing **35.31** inches

(footing 77.6" − column 7") ÷ 2 = **35.3"** — the footing projects past the column by its own thickness (45° spread).

Joist, beam and column boxes have more than one right answer — the key shows one that works. Loads use ROUNDUP like his A5 key; snow psf kept to 2 decimals.